

一、CACBA CCBAB CDAB

三、(1) $I_{p\Delta} = \frac{P}{U_p \cos \Phi} = 27.5 \text{ A}$

设 $\dot{U}_{AB} = 220 \angle 0^\circ \text{ V} \quad \varphi = 36.8^\circ$

$\dot{I}_{AB} = 27.5 \angle -36.8^\circ \text{ A} \quad \dot{I}_{BC} = 27.5 \angle -156.8^\circ \text{ A} \quad \dot{I}_{CA} = 27.5 \angle 83.2^\circ \text{ A}$

负载相电流 $I_{p\Delta} = 27.5 \text{ A}$

A 线电流 $\dot{I}_{A\Delta} = 47.6 \angle -66.8^\circ \text{ A}$

(2) 对 Y 接负载: $\dot{I}_{AY} = 12.7 \angle -30^\circ \text{ A}$

负载相电流 $I_{pY} = 12.7 \text{ A}$

A 相总线电流 $\dot{I}_A = \dot{I}_{A\Delta} + \dot{I}_{AY} = 58.29 \angle -59.2^\circ \text{ A}$

各相总的线电流为 58.29 A

四、非客观题:

三台电动机顺序起动的控制电路, M_1 起动后 M_2 才能起动, M_2 起动后 M_3 才能起动, 同时停转和分别停转。

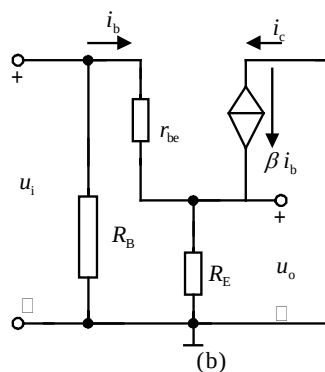
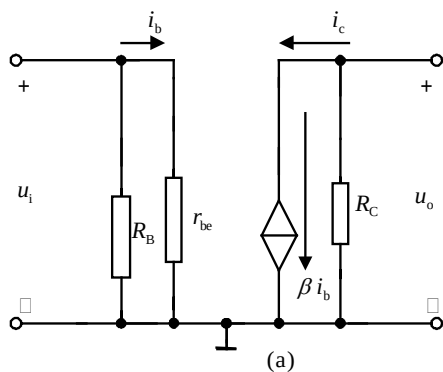
五、 $u_{O1} = -\frac{R_F}{R_1} u_{I1} = -u_{I1} = -1 \text{ V}$

$$u_O = -\frac{1}{R_2 C} \int u_{I2} dt - \frac{1}{R_3 C} \int u_{O1} dt = -\frac{1}{5 \times 10^4 \times 10^{-4}} \int 0.5 dt - \frac{1}{10^5 \times 10^{-4}} \int -1 dt$$

$$= -0.2 \int_0^{10} 0.5 dt + 0.1 \int_0^{10} 1 dt = -0.1t \Big|_0^{10} + 0.1t \Big|_0^{10} = 0$$

因此 $t = 10 \text{ s}$ 秒后 $u_O = 0$

六、(1)



(2) $A_u = 1$

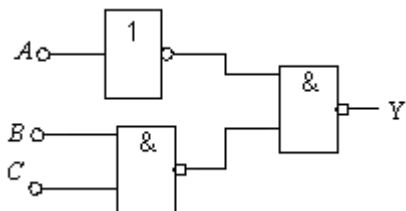
(3) 图 1 是分压式偏置放大电路; 图 2 是射级输出器

七、由题意可以得到状态真值表

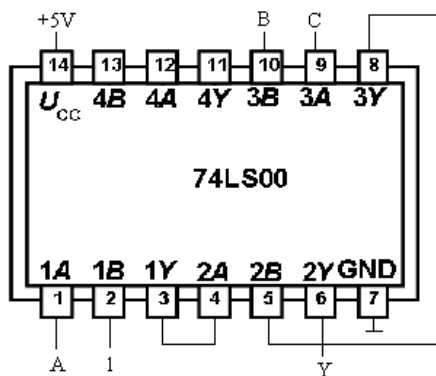
A	B	C	Y
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

可以得到表达式 $Y = A + \bar{A}BC = A + BC = \overline{\overline{A}BC}$

逻辑图为



芯片接线图如图所示



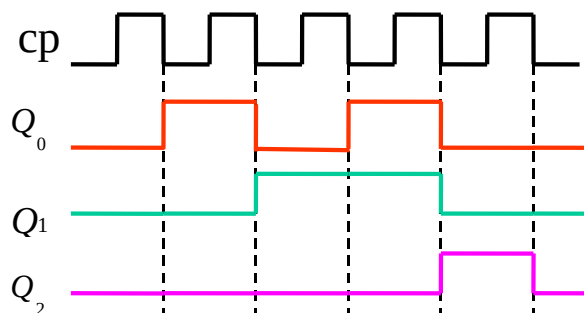
八、 (1) 输入

$$J_0 = \overline{Q_2} \quad K_0 = 1$$

$$J_1 = \overline{K_1} = \overline{Q_0}$$

$$J_2 = \overline{Q_1 + Q_0} = \overline{Q_1} \overline{Q_0} \quad K_2 = 1$$

(3) 波形图为:



(2) 状态转换表

C	Q ₂	Q ₁	Q ₀
0	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	0	0	0

此计数器为同步五进制的加法计数器